

數位電路實驗 Lab3 team9

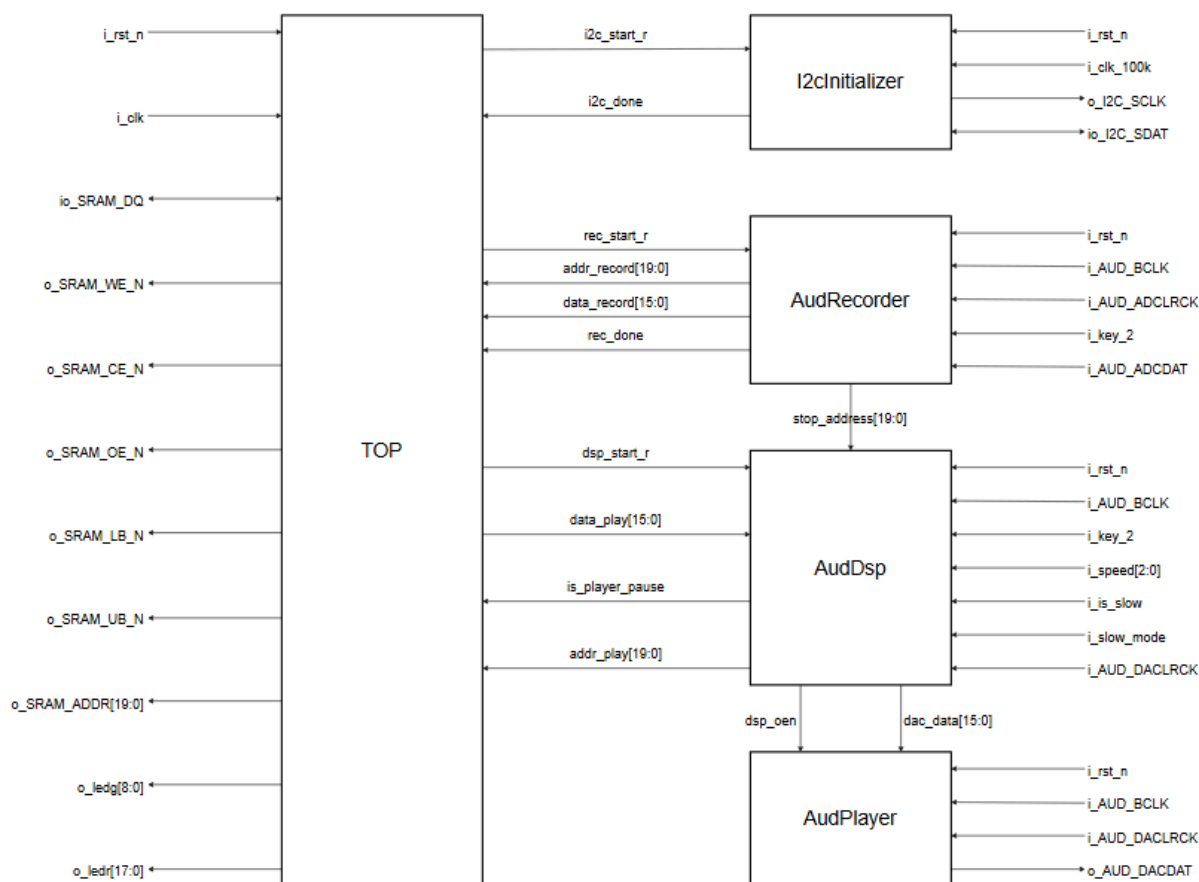
B12901005 郭力銘 B12901185 謝承恩 B12901191 彭子綸

1. File Structure

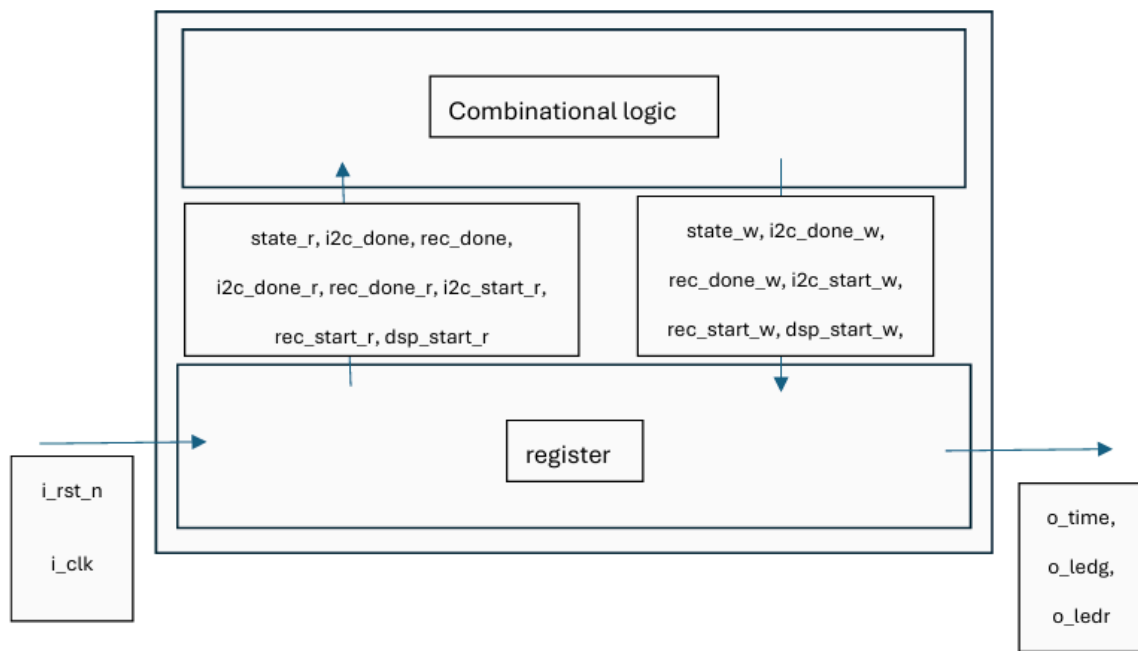
```
team09_lab3
|- team09_lab3_report
|- src
|  |- DE2_115
|     |- DE2_115.qsf
|     |- DE2_115.sdc
|     |- DE2_115.sv
|     |- Debounce.sv
|     |- SevenHexDecoder.sv
|  |- AudDSP.sv
|  |- AudPlayer.sv
|  |- AudRecorder.sv
|  |- I2cInitializer.sv
|  |- Top.sv
|  |- FastSlow.sv
```

2. System Architecture

a. Whole System



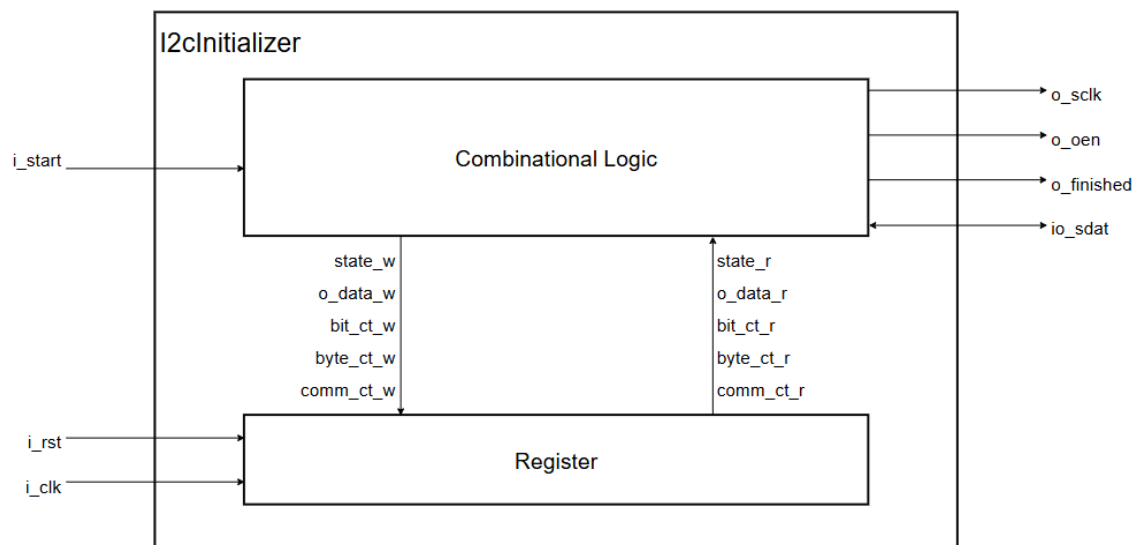
b. Top



Input/Output	Explanation
<code>i_rst_n</code>	Reset the recorder
<code>i_clk</code>	Clock
<code>state_r</code>	current state
<code>i2c_done</code>	I2C module signals finish
<code>rec_done</code>	recorder module signals finish
<code>i2c_done_r</code>	register that stand for I2C finish (if (<code>i2c_done</code>) → become 1 and hold)
<code>rec_done_r</code>	register that stand for Recorder finish (if (<code>rec_done</code>) → become 1 and hold)
<code>i2c_start_r</code>	register to make I2CInitializer start
<code>rec_start_r</code>	register to make AudRecorder start
<code>dsp_start_r</code>	register to make AudDSP start
<code>state_w</code>	next state
<code>i2c_done_w</code>	next Latched I2C finish value
<code>rec_done_w</code>	next Latched Recorder finish value
<code>i2c_start_w</code>	next signal to make I2CInitializer start (combinational)
<code>rec_start_w</code>	next signal to make AudRecorder start (combinational)

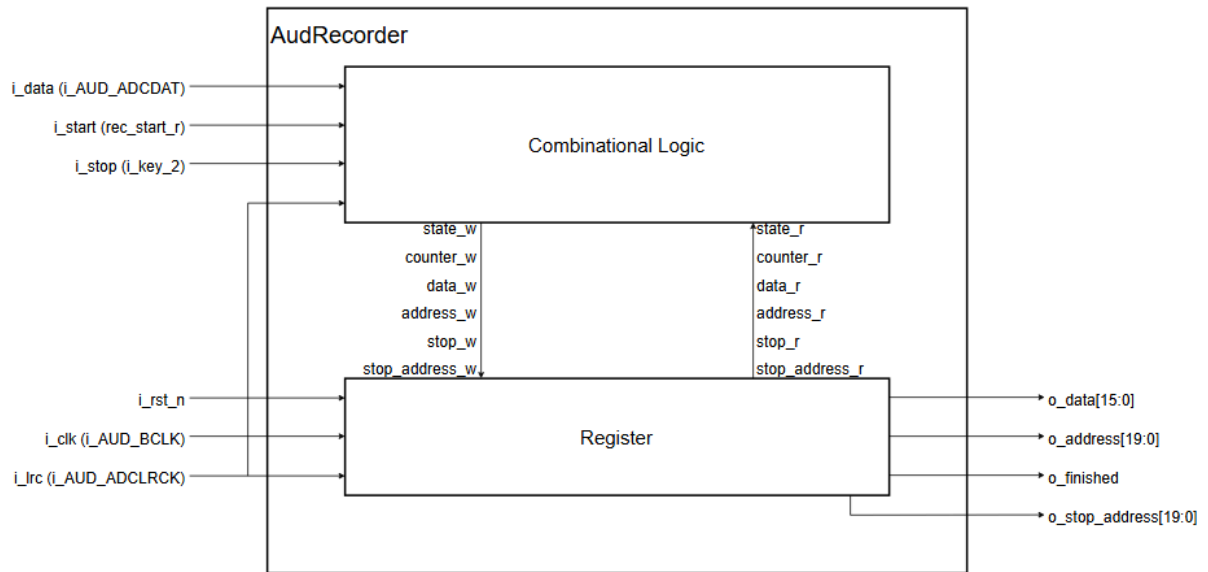
dsp_start_w	next signal to make AudDSP start (combinational)
o_time	time on FPGA
o_ledg	green led on FPGA
o_ledr	red led on FPGA

c. I2cInitializer



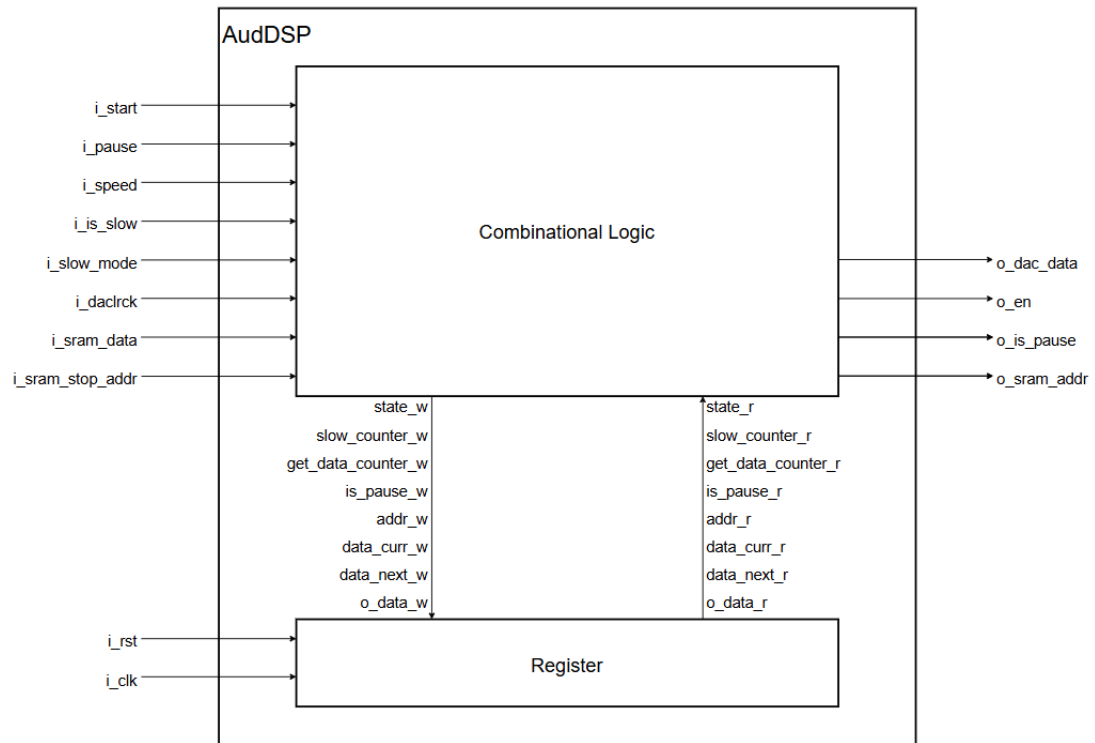
Input/Output	Explanation
i_rst	Reset the initializer.
i_clk	Clock
i_start	Start the initializer.
o_sclk	The output clk of I2C.
o_oen	1 means io_sdat is output; 0 means io_sdat is input.
o_finished	Tell the TOP module that the initializing is finished.
io_sdat	The output data and the input data from the slave.

d. AudRecorder



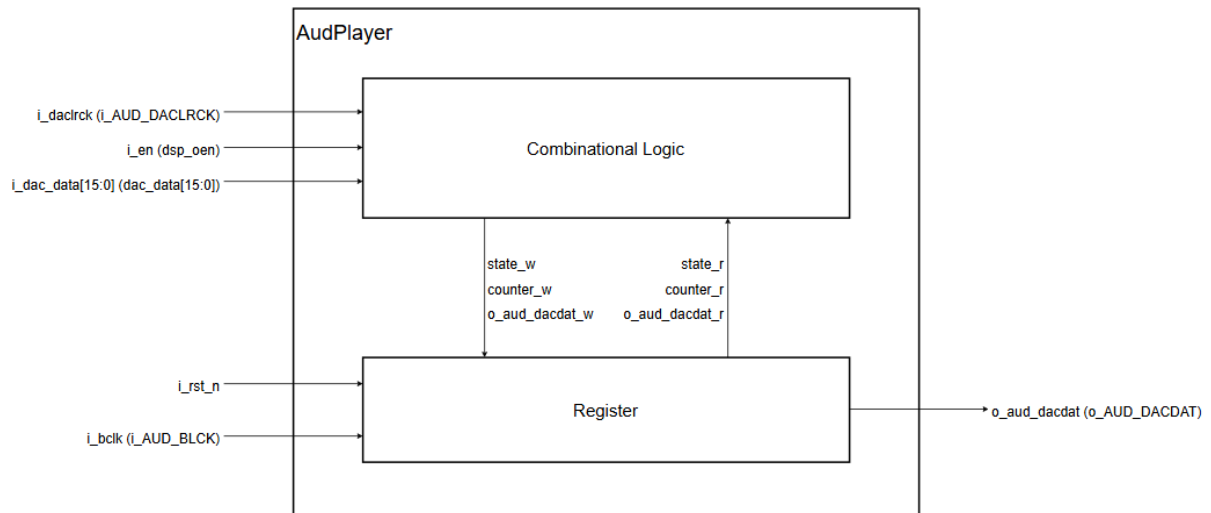
Input/Output	Explanation
i_rst_n	Reset the recorder
i_clk	Clock
i_lrc	Also clock, but represents audio channel (1 for right while 0 for left)
i_start	Check whether to start to record
i_stop	Check whether to stop to record
i_data	Data transmitted to AudRecorder
o_data[15:0]	Data to write out
o_address[19:0]	The address where the o_data should write
o_stop_address[19:0]	The address where the last data written
o_finished	Check whether AudRecorder is finished

e. AudDSP



Input/Output	Explanation
<code>i_rst</code>	Reset the DSP.
<code>i_clk</code>	Clock
<code>i_start</code>	Start the DSP.
<code>i_pause</code>	Tell the DSP module whether to pause while playbacking.
<code>i_speed</code>	Tell the DSP module the speed at which to playback.
<code>i_is_slow</code>	Tell the DSP module whether to play in slow motion.
<code>i_slow_mode</code>	Tell the DSP module which mode to use for slow playback.
<code>i_daclrck</code>	Tell the DSP module whether to get data or send data.
<code>i_sram_data</code>	The data retrieved from SRAM.
<code>i_sram_stop_addr</code>	Address of the last stop.
<code>o_dac_data</code>	The output data.
<code>o_en</code>	Indicate whether the output is enabled.
<code>o_is_pause</code>	Indicate whether it is currently in a paused state.
<code>o_sram_addr</code>	Indicates the address that is currently being output.

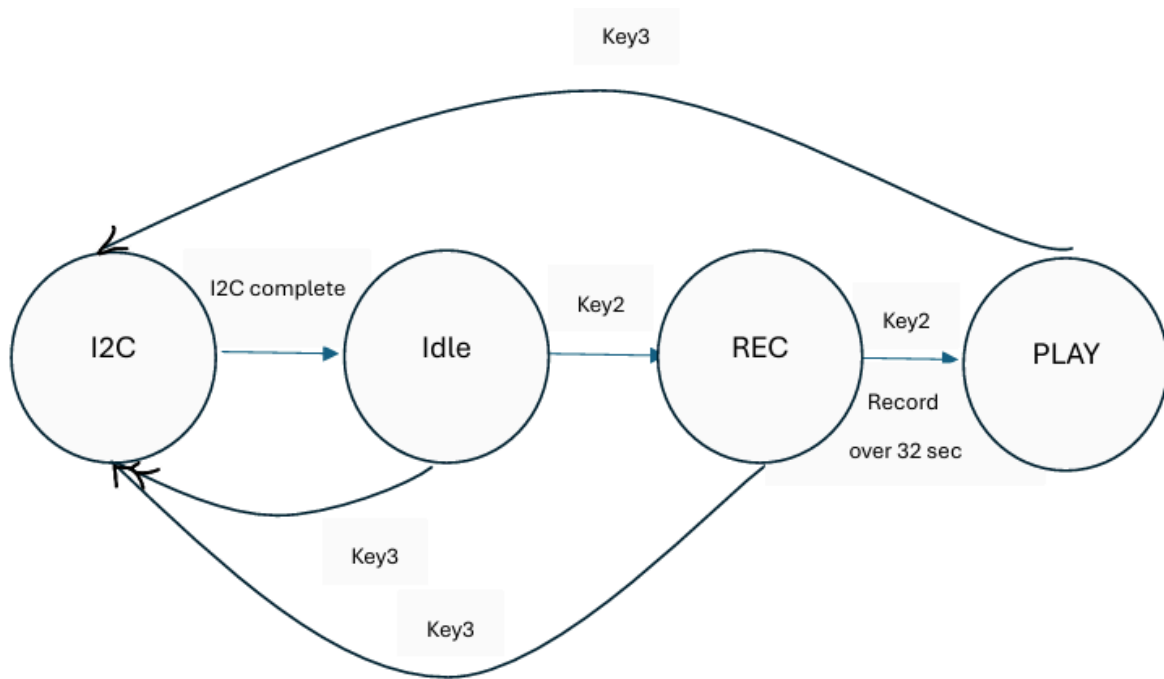
f. AudPlayer



Input/Output	Explanation
<code>i_rst_n</code>	Reset the player
<code>i_bclk</code>	Clock
<code>i_dacclk</code>	Also clock, but represents audio channel (1 for right while 0 for left)
<code>i_en</code>	Check whether it is ready to output
<code>i_dac_data[15:0]</code>	The input data caught from RAM
<code>o_aud_dacdat</code>	The output data

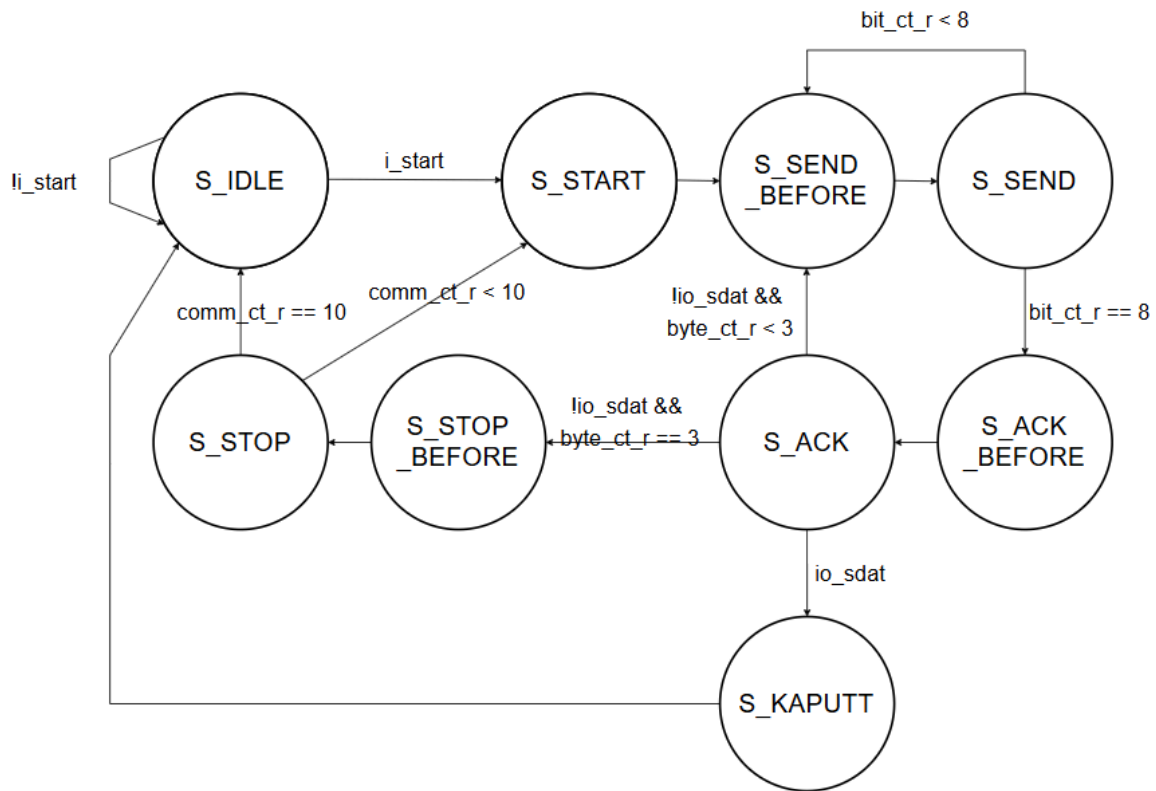
3. State Diagram & Explanation

a. Top



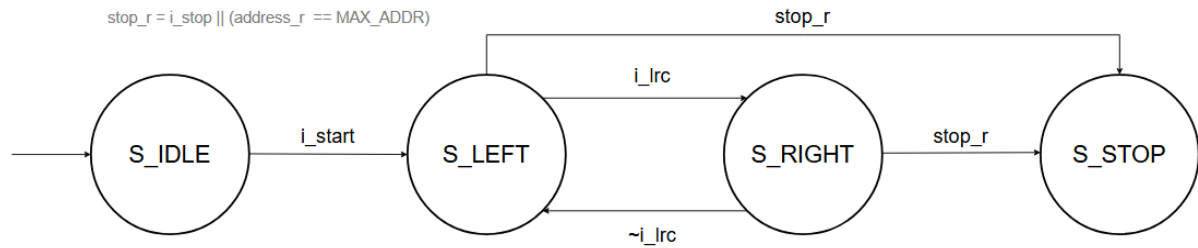
State	Explanation
I2C	initialization
Idle	Wait for key2 being pressed
REC	record the voice
PLAY	playing record/ ready to play record

b. I2CInitializer



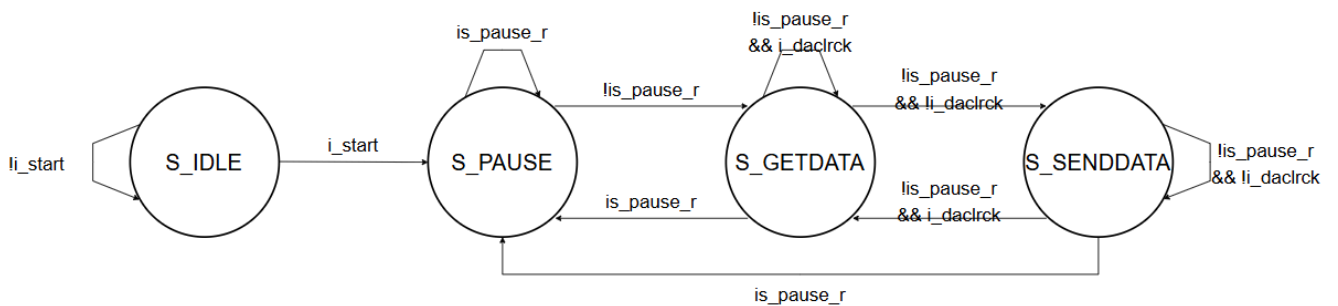
State	Explanation
S_IDLE	Wait for i_start.
S_START	Drive start condition onto the I2C bus to begin a transaction.
S_SEND_BEFORE	Prepare the next data bit for transmission (set SDA before SCL edge).
S_SEND	Toggle through clocked send cycles while shifting out data bits.
S_ACK_BEFORE	Release SDA and prepare to sample the ACK bit from the slave.
S_ACK	Sample the ACK/NACK from the slave and decide next action (continue or error).
S_STOP_BEFORE	Drive SDA low while SCL is high to prepare the stop condition sequence.
S_STOP	Generate stop condition, finalize the transaction, and advance to next command.
S_KAPUTT	Error state entered on NACK; reports failure and returns to idle.

c. AudRecorder



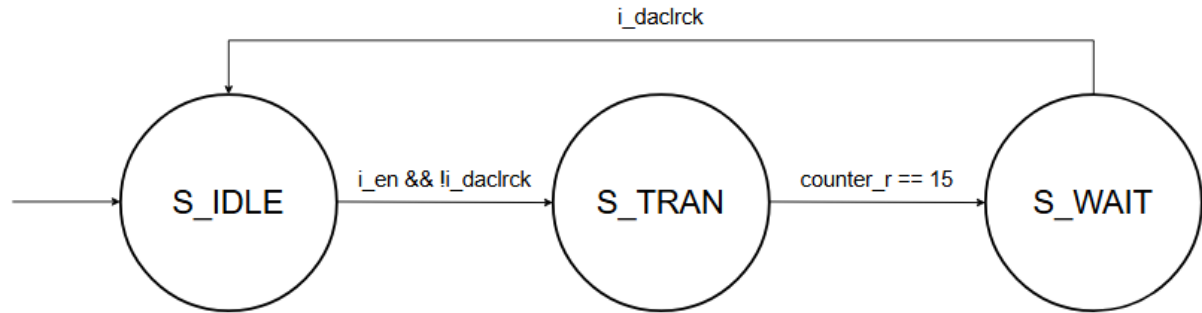
State	Explanation
S_IDLE	Wait for i_start to be pressed to start recording
S_LEFT	Left audio channel, clear the data and counter record in the last clock
S_RIGHT	Right audio channel, start to read the data then write them into RAM
S_STOP	Finish, return o_finish

d. AudDSP



State	Explanation
S_IDLE	Wait for i_start.
S_PAUSE	The module halts progress and keeps outputs/addresses stable until i_pause is released.
S_GETDATA	When i_daclrck, the module increments SRAM read address(s) and captures new sample(s) from i_sram_data across several cycles to load data_curr and data_next.
S_SENDDATA	When !i_daclrck, the module produces or holds the DAC output (including slow linear interpolation when enabled) while advancing slow counters.

e. AudPlayer



State	Explanation
S_IDLE	Wait for i_en and the audio channel to be left channel
S_TRAN	Right channel, start to play for 16 bits
S_WAIT	Already write 16 bits, wait for the audio channel becomes right channel

4. Fitter Summary

Fitter Summary	
Fitter Status	Successful - Wed Oct 29 23:56:28 2025
Quartus II 64-Bit Version	15.0.0 Build 145 04/22/2015 SJ Full Version
Revision Name	DE2_115
Top-level Entity Name	DE2_115
Family	Cyclone IV E
Device	EP4CE115F29C7
Timing Models	Final
Total logic elements	3,075 / 114,480 (3 %)
Total combinational functions	1,684 / 114,480 (1 %)
Dedicated logic registers	2,266 / 114,480 (2 %)
Total registers	2267
Total pins	518 / 529 (98 %)
Total virtual pins	0
Total memory bits	69,632 / 3,981,312 (2 %)
Embedded Multiplier 9-bit elements	2 / 532 (< 1 %)
Total PLLs	1 / 4 (25 %)

5. Timing Analyzer

Timing Closure Recommendations		
Summary [hide details]		
This design contains failing setup paths with a worst-case slack of -5.432 ns. Run Report Timing Closure Recommendations for recommendations on how to close setup timing. For recommendations for any particular path, click the appropriate link in the table below.		
Top Failing Paths [hide details]		
Slack	From	To
1 -5.432	Top:top0[AudRecor...r]state_r:S_STOP	Top:top0[state_r:S_REC
2 -4.763	Top:top0[AudRecor...er]address_r[17]	sls_signaltap:aut...data_in_reg[124]
3 -4.762	Top:top0[AudRecor...er]address_r[17]	sls_signaltap:aut...data_in_reg[78]
4 -4.751	Top:top0[AudSP:dep]addr_r[16]	sls_signaltap:aut...gger_in_reg[77]
5 -4.660	Top:top0[AudRecor...er]address_r[19]	sls_signaltap:aut...gger_in_reg[80]

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Fmax Summary

Timing Closure Recomm

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Removal Summary

Minimum Delay Width Sum

	Clock	Slack	End Point TNS
1	pll0 altpll_0 sd1 pll7 clk[0]	-5.432	-246.047
2	AUD_BCLK	-2.535	-53.456
3	altera_reserved_tck	43.021	0.000
4	pll0 altpll_0 sd1 pll7 clk[1]	78.961	0.000

Top.sv

Compilation Report - DE2_115

AudRecorder.sv

AudPlayer.sv

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Worst-Case Timing Paths

Datasheet Report

	Clock	Slack	End Point TNS
1	pll0 altpll_0 sd1 pll7 clk[0]	-0.371	-0.738
2	altera_reserved_tck	0.402	0.000
3	pll0 altpll_0 sd1 pll7 clk[1]	0.405	0.000
4	AUD_BCLK	0.440	0.000

Top.sv		Compilation Report - DE2_115		AudRecorder.sv		AudPlayer.sv			
Table of Contents		Slow 1200mV BSC Model Setup: p0[altpl_0]sd1[p17]clk[0]							
		Slack	From Node	To Node	Launch Clock	Latch Clock	Relationship	Clock Skew	
Flow Summary		1	-4.532	Top:top0[AudRecorder:recorder]state_r[5_STOP	Top:top0[state_r[5_REC	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.726
Flow Settings		2	-4.763	Top:top0[AudRecorder:recorder]address_r[17]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[124]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.983
Flow Non-Default Global Settings		3	-4.762	Top:top0[AudRecorder:recorder]address_r[17]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[78]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.983
Flow Elapsed Time		4	-4.751	Top:top0[AudDSP:dsp]addr_r[16]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[77]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.979
Flow OS Summary		5	-4.660	Top:top0[AudRecorder:recorder]address_r[19]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[80]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.982
Flow Log		6	-4.659	Top:top0[AudRecorder:recorder]address_r[19]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[80]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.982
Analysis & Synthesis		7	-4.658	Top:top0[AudRecorder:recorder]address_r[19]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[126]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.982
Fitter		8	-4.653	Top:top0[AudRecorder:recorder]address_r[19]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[126]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.982
Flow Messages		9	-4.621	Top:top0[AudRecorder:recorder]address_r[18]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[79]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.982
Flow Suppressed Messages		10	-4.620	Top:top0[AudRecorder:recorder]address_r[18]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[125]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.982
Assembler		11	-4.620	Top:top0[AudRecorder:recorder]address_r[18]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[79]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.982
TimeQuest Timing Analyzer		12	-4.620	Top:top0[AudRecorder:recorder]address_r[18]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[125]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.982
Summary		13	-4.599	Top:top0[AudDSP:dsp]addr_r[17]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[78]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.983
Parallel Compilation		14	-4.598	Top:top0[AudDSP:dsp]addr_r[17]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[78]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.983
SDC File List		15	-4.534	Top:top0[AudDSP:dsp]addr_r[19]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[80]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.907
Clocks		16	-4.533	Top:top0[AudDSP:dsp]addr_r[19]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[80]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.907
Slow 1200mV BSC Model		17	-4.532	Top:top0[AudDSP:dsp]addr_r[19]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[126]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.907
Fmax Summary		18	-4.528	Top:top0[AudDSP:dsp]addr_r[18]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[79]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.907
Timing Closure Recommendations		19	-4.527	Top:top0[AudDSP:dsp]addr_r[19]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[126]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.907
Setup Summary		20	-4.527	Top:top0[AudDSP:dsp]addr_r[18]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[125]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.907
Hold Summary		21	-4.527	Top:top0[AudDSP:dsp]addr_r[18]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[79]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.907
Recovery Summary		22	-4.527	Top:top0[AudDSP:dsp]addr_r[18]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[125]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.907
Removal Summary		23	-4.519	Top:top0[AudRecorder:recorder]address_r[13]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[74]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.624
Minimum Pulse Width Summary		24	-4.519	Top:top0[AudRecorder:recorder]address_r[13]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[74]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.624
Worst-Case Timing Paths		25	-4.510	Top:top0[AudRecorder:recorder]address_r[6]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[86]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.707
Setup: 'p0[altpl_0]sd1[p17]clk[1]'		26	-4.508	Top:top0[AudRecorder:recorder]address_r[6]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[86]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.707
Setup: 'AUD_BCLK'		27	-4.429	Top:top0[AudRecorder:recorder]address_r[7]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[87]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.694
Hold: 'p0[altpl_0]sd1[p17]clk[0]'		28	-4.429	Top:top0[AudRecorder:recorder]address_r[7]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[87]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.694
Hold: 'altera_reserved_tck'		29	-4.411	Top:top0[AudRecorder:recorder]address_r[16]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[77]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.979
Hold: 'p0[altpl_0]sd1[p17]clk[1]'		30	-4.374	Top:top0[AudDSP:dsp]addr_r[16]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[123]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.565
Hold: 'AUD_BCLK'		31	-4.374	Top:top0[AudDSP:dsp]addr_r[16]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[77]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.565
Recovery: 'altera_reserved_tck'		32	-4.371	Top:top0[AudDSP:dsp]addr_r[16]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[123]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.565
Removal: 'altera_reserved_tck'		33	-4.345	Top:top0[AudRecorder:recorder]address_r[11]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[72]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.598
Minimum Pulse Width: 'CLOCK_50'		34	-4.344	Top:top0[AudRecorder:recorder]address_r[11]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[72]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.598
Minimum Pulse Width: 'CLOCK2_50'		35	-4.268	Top:top0[AudDSP:dsp]addr_r[12]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[73]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.847
Minimum Pulse Width: 'AUD_BCLK'		36	-4.268	Top:top0[AudDSP:dsp]addr_r[16]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[86]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.847
Minimum Pulse Width: 'altera_reserved_tck'		37	-4.266	Top:top0[AudDSP:dsp]addr_r[12]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[73]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.847
Minimum Pulse Width: 'p0[altpl_0]sd1[p17]clk[1]'		38	-4.266	Top:top0[AudDSP:dsp]addr_r[15]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[86]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.847
Minimum Pulse Width: 'AUD_BCLK'		39	-4.243	Top:top0[AudRecorder:recorder]address_r[10]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[71]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.575
Minimum Pulse Width: 'altera_reserved_tck'		40	-4.239	Top:top0[AudDSP:dsp]addr_r[0]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[70]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.818
Minimum Pulse Width: 'p0[altpl_0]sd1[p17]clk[0]'		41	-4.238	Top:top0[AudDSP:dsp]addr_r[0]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[70]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.818
Minimum Pulse Width: 'CLOCK_50'		42	-4.237	Top:top0[AudRecorder:recorder]address_r[10]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[71]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.575
Minimum Pulse Width: 'CLOCK2_50'		43	-4.236	Top:top0[AudDSP:dsp]addr_r[5]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[85]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.820
Minimum Pulse Width: 'AUD_BCLK'		44	-4.236	Top:top0[AudDSP:dsp]addr_r[5]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[85]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.820
Minimum Pulse Width: 'altera_reserved_tck'		45	-4.236	Top:top0[AudDSP:dsp]addr_r[7]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[87]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.818
Minimum Pulse Width: 'p0[altpl_0]sd1[p17]clk[0]'		46	-4.236	Top:top0[AudDSP:dsp]addr_r[7]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[87]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.818
Minimum Pulse Width: 'AUD_BCLK'		47	-4.217	Top:top0[AudDSP:dsp]addr_r[14]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[75]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.776
Minimum Pulse Width: 'altera_reserved_tck'		48	-4.217	Top:top0[AudDSP:dsp]addr_r[14]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[75]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.776
Minimum Pulse Width: 'p0[altpl_0]sd1[p17]clk[1]'		49	-4.211	Top:top0[AudDSP:dsp]addr_r[115]	sid_signaltap:auto_signaltap_0[acq_data_in_reg[122]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.772
Minimum Pulse Width: 'AUD_BCLK'		50	-4.210	Top:top0[AudDSP:dsp]addr_r[115]	sid_signaltap:auto_signaltap_0[acq_trigger_in_reg[122]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.772

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		Slack	From Node	To Node	Launch Clock	Latch Clock	Relationship	Clock Skew
Flow Summary		51 -4.209	Top:top0[AudDSP:dsp]addr_r[15]	sid_signaltap:aut_0[acq_data_in_reg[76]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.772
Flow Settings		52 -4.209	Top:top0[AudDSP:dsp]addr_r[8]	sid_signaltap:aut_0[acq_data_in_reg[88]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.789
Flow Non-Default Global Settings		53 -4.207	Top:top0[AudDSP:dsp]addr_r[15]	sid_signaltap:aut_0[acq_trigger_in_reg[76]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.772
Flow Elapsed Time		54 -4.207	Top:top0[AudDSP:dsp]addr_r[8]	sid_signaltap:aut_0[acq_trigger_in_reg[88]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.789
Flow OS Summary		55 -4.176	Top:top0[AudRecorder:recorder]address_r[17]	sid_signaltap:aut_0[acq_trigger_in_reg[78]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.565
Flow Log		56 -4.171	Top:top0[AudRecorder:recorder]address_r[17]	sid_signaltap:aut_0[acq_trigger_in_reg[124]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.565
Analysis & Synthesis		57 -4.107	Top:top0[AudRecorder:recorder]address_r[12]	sid_signaltap:aut_0[acq_trigger_in_reg[73]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.847
Fitter		58 -4.105	Top:top0[AudRecorder:recorder]address_r[12]	sid_signaltap:aut_0[acq_data_in_reg[73]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.847
Flow Messages		59 -4.077	Top:top0[AudRecorder:recorder]address_r[0]	sid_signaltap:aut_0[acq_trigger_in_reg[70]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.818
Flow Suppressed Messages		60 -4.077	Top:top0[AudRecorder:recorder]address_r[5]	sid_signaltap:aut_0[acq_trigger_in_reg[85]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.820
Assembler		62 -4.076	Top:top0[AudRecorder:recorder]address_r[0]	sid_signaltap:aut_0[acq_data_in_reg[70]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.818
TimeQuest Timing Analyzer		63 -4.069	Top:top0[AudRecorder:recorder]address_r[8]	sid_signaltap:aut_0[acq_data_in_reg[88]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.789
Summary		64 -4.067	Top:top0[AudRecorder:recorder]address_r[8]	sid_signaltap:aut_0[acq_trigger_in_reg[88]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.789
Parallel Compilation		65 -4.055	Top:top0[AudRecorder:recorder]address_r[14]	sid_signaltap:aut_0[acq_trigger_in_reg[75]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.776
SDC File List		66 -4.055	Top:top0[AudRecorder:recorder]address_r[14]	sid_signaltap:aut_0[acq_data_in_reg[75]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.776
Clocks		67 -4.050	Top:top0[AudRecorder:recorder]address_r[15]	sid_signaltap:aut_0[acq_data_in_reg[122]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.772
Slow 1200mV BSC Model		68 -4.049	Top:top0[AudRecorder:recorder]address_r[15]	sid_signaltap:aut_0[acq_trigger_in_reg[122]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.772
Fmax Summary		69 -4.048	Top:top0[AudRecorder:recorder]address_r[15]	sid_signaltap:aut_0[acq_data_in_reg[76]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.772
Timing Closure Recommendations		70 -4.034	Top:top0[AudRecorder:recorder]address_r[16]	sid_signaltap:aut_0[acq_trigger_in_reg[123]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.565
Setup Summary		71 -4.034	Top:top0[AudRecorder:recorder]address_r[16]	sid_signaltap:aut_0[acq_data_in_reg[77]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.565
Hold Summary		72 -4.031	Top:top0[AudRecorder:recorder]address_r[16]	sid_signaltap:aut_0[acq_data_in_reg[123]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.565
Recovery Summary		73 -4.030	Top:top0[AudRecorder:recorder]address_r[9]	sid_signaltap:aut_0[acq_trigger_in_reg[89]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.323
Removal Summary		74 -4.030	Top:top0[AudRecorder:recorder]address_r[9]	sid_signaltap:aut_0[acq_data_in_reg[89]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.323
Minimum Pulse Width Summary		75 -4.017	Top:top0[AudDSP:dsp]pause_r	sid_signaltap:aut_0[acq_trigger_in_reg[110]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.762
Worst-Case Timing Paths		76 -4.016	Top:top0[AudDSP:dsp]pause_r	sid_signaltap:aut_0[acq_data_in_reg[110]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.762
	Setup: p0[altpl_0]sd1[p17]clk[0]	77 -4.012	Top:top0[AudDSP:dsp]addr_r[17]	sid_signaltap:aut_0[acq_trigger_in_reg[78]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.565
	Setup: AUD_BCLK	78 -4.007	Top:top0[AudRecorder:recorder]state_r[5_STOP]	Top:top0[state_r[5_PLAY]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.776
	Setup: altera_reserved_tck	81 -3.923	Top:top0[AudDSP:dsp]addr_r[9]	sid_signaltap:aut_0[acq_trigger_in_reg[89]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.494
	Hold: p0[altpl_0]sd1[p17]clk[1]	82 -3.923	Top:top0[AudDSP:dsp]addr_r[9]	sid_signaltap:aut_0[acq_data_in_reg[89]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.494
	Hold: p0[altpl_0]sd1[p17]clk[1]	83 -3.913	Top:top0[AudDSP:dsp]addr_r[2]	sid_signaltap:aut_0[acq_data_in_reg[82]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.314
	Hold: p0[altpl_0]sd1[p17]clk[1]	84 -3.912	Top:top0[AudDSP:dsp]addr_r[1]	sid_signaltap:aut_0[acq_trigger_in_reg[81]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.314
	Hold: altera_reserved_tck	85 -3.912	Top:top0[AudDSP:dsp]addr_r[2]	sid_signaltap:aut_0[acq_trigger_in_reg[82]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.314
	Hold: p0[altpl_0]sd1[p17]clk[1]	86 -3.911	Top:top0[AudDSP:dsp]addr_r[1]	sid_signaltap:aut_0[acq_data_in_reg[81]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.314
	Hold: AUD_BCLK	87 -3.909	Top:top0[AudDSP:dsp]addr_r[10]	sid_signaltap:aut_0[acq_data_in_reg[71]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.314
	Recovery: altera_reserved_tck	88 -3.903	Top:top0[AudDSP:dsp]addr_r[10]	sid_signaltap:aut_0[acq_trigger_in_reg[71]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.314
	Removal: altera_reserved_tck	89 -3.889	Top:top0[AudRecorder:recorder]address_r[4]	sid_signaltap:aut_0[acq_trigger_in_reg[84]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.297
	Minimum Pulse Width: CLOCK_50	90 -3.886	Top:top0[AudRecorder:recorder]address_r[4]	sid_signaltap:aut_0[acq_data_in_reg[84]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.297
	Minimum Pulse Width: CLOCK2_50	91 -3.883	Top:top0[AudRecorder:recorder]address_r[1]	sid_signaltap:aut_0[acq_trigger_in_reg[81]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.274
	Minimum Pulse Width: CLOCK_50	92 -3.882	Top:top0[AudRecorder:recorder]address_r[1]	sid_signaltap:aut_0[acq_data_in_reg[81]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.274
	Minimum Pulse Width: CLOCK2_50	93 -3.866	Top:top0[AudRecorder:recorder]address_r[2]	sid_signaltap:aut_0[acq_data_in_reg[82]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.274
	Minimum Pulse Width: AUD_BCLK	94 -3.865	Top:top0[AudRecorder:recorder]address_r[2]	sid_signaltap:aut_0[acq_trigger_in_reg[82]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.274
	Minimum Pulse Width: p0[altpl_0]sd1[p17]clk[0]	95 -3.763	Top:top0[AudDSP:dsp]addr_r[13]	sid_signaltap:aut_0[acq_trigger_in_reg[74]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.494
	Minimum Pulse Width: altera_reserved_tck	96 -3.763	Top:top0[AudDSP:dsp]addr_r[13]	sid_signaltap:aut_0[acq_data_in_reg[74]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.494
	Minimum Pulse Width: p0[altpl_0]sd1[p17]clk[1]	97 -3.754	Top:top0[AudRecorder:recorder]address_r[3]	sid_signaltap:aut_0[acq_data_in_reg[83]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.297
	Minimum Pulse Width: p0[altpl_0]sd1[p17]clk[0]	98 -3.753	Top:top0[AudRecorder:recorder]address_r[3]	sid_signaltap:aut_0[acq_trigger_in_reg[83]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.297
	Dataset Report	99 -6.689	Top:top0[AudRecorder:recorder]state_r[5_STOP]	Top:top0[rec_done_r]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.796
	Assembly	100 -3.522	Top:top0[AudRecorder:dsp]addr_r[3]	sid_signaltap:aut_0[acq_data_in_reg[83]	AUD_BCLK	p0[altpl_0]sd1[p17]clk[0]	0.001	-2.091

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Datasheet Report

Slow 120nmv BSC Model Setup: 'AUD_BCLK'

	Slack	From Node	To Node	Launch Clock	Latch Clock	Relationship	Clock Skew	Data Delay
51	17.544	Top:top0[AudDSP:dsdp]data_next_r[2]	Top:top0[AudDSP:dsdp]_o_data_r[10]	AUD_BCLK	AUD_BCLK	83.000	0.043	65.517
52	17.546	Top:top0[AudDSP:dsdp]data_curr_r[1]	Top:top0[AudDSP:dsdp]_o_data_r[11]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.286
53	17.557	Top:top0[AudDSP:dsdp]data_curr_r[6]	Top:top0[AudDSP:dsdp]_o_data_r[14]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.275
54	17.566	Top:top0[AudDSP:dsdp]data_curr_r[0]	Top:top0[AudDSP:dsdp]_o_data_r[11]	AUD_BCLK	AUD_BCLK	83.000	-0.084	65.368
55	17.574	Top:top0[AudDSP:dsdp]data_next_r[2]	Top:top0[AudDSP:dsdp]_o_data_r[8]	AUD_BCLK	AUD_BCLK	83.000	0.043	65.467
56	17.578	Top:top0[AudDSP:dsdp]data_curr_r[8]	Top:top0[AudDSP:dsdp]_o_data_r[15]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.264
57	17.582	Top:top0[AudDSP:dsdp]data_curr_r[3]	Top:top0[AudDSP:dsdp]_o_data_r[9]	AUD_BCLK	AUD_BCLK	83.000	0.043	65.479
58	17.606	Top:top0[AudDSP:dsdp]data_curr_r[4]	Top:top0[AudDSP:dsdp]_o_data_r[15]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.226
59	17.613	Top:top0[AudDSP:dsdp]data_curr_r[3]	Top:top0[AudDSP:dsdp]_o_data_r[12]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.219
60	17.627	Top:top0[AudDSP:dsdp]data_next_r[0]	Top:top0[AudDSP:dsdp]_o_data_r[4]	AUD_BCLK	AUD_BCLK	83.000	0.043	65.434
61	17.628	Top:top0[AudDSP:dsdp]data_curr_r[2]	Top:top0[AudDSP:dsdp]_o_data_r[11]	AUD_BCLK	AUD_BCLK	83.000	-0.084	65.306
62	17.629	Top:top0[AudDSP:dsdp]data_next_r[0]	Top:top0[AudDSP:dsdp]_o_data_r[5]	AUD_BCLK	AUD_BCLK	83.000	0.043	65.432
63	17.636	Top:top0[AudDSP:dsdp]data_curr_r[5]	Top:top0[AudDSP:dsdp]_o_data_r[14]	AUD_BCLK	AUD_BCLK	83.000	-0.084	65.298
64	17.641	Top:top0[AudDSP:dsdp]data_curr_r[4]	Top:top0[AudDSP:dsdp]_o_data_r[12]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.191
65	17.642	Top:top0[AudDSP:dsdp]data_next_r[2]	Top:top0[AudDSP:dsdp]_o_data_r[7]	AUD_BCLK	AUD_BCLK	83.000	0.043	65.419
66	17.644	Top:top0[AudDSP:dsdp]data_curr_r[1]	Top:top0[AudDSP:dsdp]_o_data_r[13]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.168
67	17.660	Top:top0[AudDSP:dsdp]data_next_r[2]	Top:top0[AudDSP:dsdp]_o_data_r[6]	AUD_BCLK	AUD_BCLK	83.000	0.043	65.393
68	17.678	Top:top0[AudDSP:dsdp]data_curr_r[6]	Top:top0[AudDSP:dsdp]_o_data_r[15]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.194
69	17.683	Top:top0[AudDSP:dsdp]data_curr_r[1]	Top:top0[AudDSP:dsdp]_o_data_r[10]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.149
70	17.684	Top:top0[AudDSP:dsdp]data_curr_r[0]	Top:top0[AudDSP:dsdp]_o_data_r[13]	AUD_BCLK	AUD_BCLK	83.000	-0.084	65.250
71	17.703	Top:top0[AudDSP:dsdp]data_curr_r[7]	Top:top0[AudDSP:dsdp]_o_data_r[14]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.129
72	17.703	Top:top0[AudDSP:dsdp]data_curr_r[0]	Top:top0[AudDSP:dsdp]_o_data_r[10]	AUD_BCLK	AUD_BCLK	83.000	-0.084	65.231
73	17.713	Top:top0[AudDSP:dsdp]data_curr_r[1]	Top:top0[AudDSP:dsdp]_o_data_r[8]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.119
74	17.713	Top:top0[AudDSP:dsdp]data_curr_r[6]	Top:top0[AudDSP:dsdp]_o_data_r[12]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.119
75	17.726	Top:top0[AudDSP:dsdp]data_curr_r[3]	Top:top0[AudDSP:dsdp]_o_data_r[11]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.106
76	17.733	Top:top0[AudDSP:dsdp]data_curr_r[0]	Top:top0[AudDSP:dsdp]_o_data_r[8]	AUD_BCLK	AUD_BCLK	83.000	-0.084	65.201
77	17.741	Top:top0[AudDSP:dsdp]data_next_r[0]	Top:top0[AudDSP:dsdp]_o_data_r[2]	AUD_BCLK	AUD_BCLK	83.000	0.043	65.320
78	17.746	Top:top0[AudDSP:dsdp]data_curr_r[2]	Top:top0[AudDSP:dsdp]_o_data_r[13]	AUD_BCLK	AUD_BCLK	83.000	-0.084	65.188
79	17.754	Top:top0[AudDSP:dsdp]data_next_r[2]	Top:top0[AudDSP:dsdp]_o_data_r[13]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.076
80	17.757	Top:top0[AudDSP:dsdp]data_curr_r[5]	Top:top0[AudDSP:dsdp]_o_data_r[15]	AUD_BCLK	AUD_BCLK	83.000	-0.084	65.177
81	17.765	Top:top0[AudDSP:dsdp]data_curr_r[2]	Top:top0[AudDSP:dsdp]_o_data_r[10]	AUD_BCLK	AUD_BCLK	83.000	-0.084	65.169
82	17.781	Top:top0[AudDSP:dsdp]data_curr_r[1]	Top:top0[AudDSP:dsdp]_o_data_r[7]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.051
83	17.783	Top:top0[AudDSP:dsdp]data_next_r[13]	Top:top0[AudDSP:dsdp]_o_data_r[14]	AUD_BCLK	AUD_BCLK	83.000	0.043	65.278
84	17.791	Top:top0[AudDSP:dsdp]data_next_r[2]	Top:top0[AudDSP:dsdp]_o_data_r[9]	AUD_BCLK	AUD_BCLK	83.000	0.043	65.270
85	17.792	Top:top0[AudDSP:dsdp]data_curr_r[12]	Top:top0[AudDSP:dsdp]_o_data_r[12]	AUD_BCLK	AUD_BCLK	83.000	-0.084	65.142
86	17.795	Top:top0[AudDSP:dsdp]data_curr_r[8]	Top:top0[AudDSP:dsdp]_o_data_r[8]	AUD_BCLK	AUD_BCLK	83.000	-0.084	65.139
87	17.801	Top:top0[AudDSP:dsdp]data_curr_r[0]	Top:top0[AudDSP:dsdp]_o_data_r[7]	AUD_BCLK	AUD_BCLK	83.000	-0.084	65.133
88	17.807	Top:top0[AudDSP:dsdp]data_curr_r[1]	Top:top0[AudDSP:dsdp]_o_data_r[6]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.025
89	17.824	Top:top0[AudDSP:dsdp]data_curr_r[7]	Top:top0[AudDSP:dsdp]_o_data_r[15]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.008
90	17.826	Top:top0[AudDSP:dsdp]data_curr_r[6]	Top:top0[AudDSP:dsdp]_o_data_r[11]	AUD_BCLK	AUD_BCLK	83.000	-0.186	65.006
91	17.827	Top:top0[AudDSP:dsdp]data_next_r[9]	Top:top0[AudDSP:dsdp]_o_data_r[14]	AUD_BCLK	AUD_BCLK	83.000	-0.143	65.043
92	17.827	Top:top0[AudDSP:dsdp]data_curr_r[0]	Top:top0[AudDSP:dsdp]_o_data_r[6]	AUD_BCLK	AUD_BCLK	83.000	-0.084	65.107
93	17.836	Top:top0[AudDSP:dsdp]data_next_r[2]	Top:top0[AudDSP:dsdp]_o_data_r[4]	AUD_BCLK	AUD_BCLK	83.000	0.043	65.225
94	17.838	Top:top0[AudDSP:dsdp]data_next_r[2]	Top:top0[AudDSP:dsdp]_o_data_r[5]	AUD_BCLK	AUD_BCLK	83.000	0.043	65.223
95	17.844	Top:top0[AudDSP:dsdp]data_curr_r[3]	Top:top0[AudDSP:dsdp]_o_data_r[13]	AUD_BCLK	AUD_BCLK	83.000	-0.186	64.988
96	17.845	Top:top0[AudDSP:dsdp]data_next_r[8]	Top:top0[AudDSP:dsdp]_o_data_r[14]	AUD_BCLK	AUD_BCLK	83.000	-0.143	65.030
97	17.859	Top:top0[AudDSP:dsdp]data_curr_r[7]	Top:top0[AudDSP:dsdp]_o_data_r[12]	AUD_BCLK	AUD_BCLK	83.000	-0.186	64.973
98	17.863	Top:top0[AudDSP:dsdp]data_curr_r[3]	Top:top0[AudDSP:dsdp]_o_data_r[10]	AUD_BCLK	AUD_BCLK	83.000	-0.186	64.969
99	17.863	Top:top0[AudDSP:dsdp]data_curr_r[2]	Top:top0[AudDSP:dsdp]_o_data_r[7]	AUD_BCLK	AUD_BCLK	83.000	-0.084	65.071
100	17.872	Top:top0[AudDSP:dsdp]data_curr_r[4]	Top:top0[AudDSP:dsdp]_o_data_r[13]	AUD_BCLK	AUD_BCLK	83.000	-0.186	64.960

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1	p[0]a[tpbl_0]sd1[p[7]ck[0]	-4.887	-221.620
2	AUD_BCLK	-2.451	-50.218
3	altera_reserved_tck	43.715	0.000
4	p[0]a[tpbl_0]sd1[p[7]ck[1]	79.274	0.000

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Slow 1200mV 0C Model Hold Summary

	Clock	Slack	End Point TNS
1	p[0]a[tpbl_0]sd1[p[7]ck[0]	-0.354	-0.701
2	altera_reserved_tck	0.354	0.000
3	p[0]a[tpbl_0]sd1[p[7]ck[1]	0.356	0.000
4	AUD_BCLK	0.387	0.000

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Slow 1200mV 85C Model										17 -4.074 TopTopt0[AudDSP:dsp]addr_r[19]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[80]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.608																			
Slow 1200mV 0C Model										18 -4.073 TopTopt0[AudDSP:dsp]addr_r[19]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[126]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.608																			
Fmax Summary										19 -4.072 TopTopt0[AudRecorder:recorder]address_r[13]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[74]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.357																			
Setup Summary										20 -4.072 TopTopt0[AudRecorder:recorder]address_r[13]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[74]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.357																			
Hold Summary										21 -4.072 TopTopt0[AudRecorder:recorder]address_r[6]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[86]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.424																			
Recovery Summary										22 -4.069 TopTopt0[AudDSP:dsp]addr_r[18]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[79]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.608																			
Removal Summary										23 -4.069 TopTopt0[AudDSP:dsp]addr_r[18]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[79]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.608																			
Minimum Pulse Width Summary										24 -4.068 TopTopt0[AudDSP:dsp]addr_r[19]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[126]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.608																			
Worst-Case Timing Paths										25 -4.068 TopTopt0[AudDSP:dsp]addr_r[18]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[125]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.608																			
Setup: 'p0[altpl_0]sd1[p07]clk[0]'										26 -4.068 TopTopt0[AudDSP:dsp]addr_r[18]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[125]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.608																			
Setup: 'alt'era_reserved_tck'										27 -3.999 TopTopt0[AudRecorder:recorder]address_r[7]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[87]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.410																			
Setup: 'p0[altpl_0]sd1[p07]clk[1]'										28 -3.998 TopTopt0[AudRecorder:recorder]address_r[7]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[87]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.410																			
Setup: 'alt'era_reserved_tck'										29 -3.973 TopTopt0[AudRecorder:recorder]address_r[6]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[77]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.677																			
Setup: 'AUD_BCLK'										30 -3.933 TopTopt0[AudRecorder:recorder]address_r[11]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[71]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.330																			
Setup: 'alt'era_reserved_tck'										31 -3.932 TopTopt0[AudRecorder:recorder]address_r[11]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[72]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.330																			
Setup: 'p0[altpl_0]sd1[p07]clk[1]'										32 -3.918 TopTopt0[AudDSP:dsp]addr_r[16]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[123]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.295																			
Setup: 'p0[altpl_0]sd1[p07]clk[0]'										33 -3.918 TopTopt0[AudDSP:dsp]addr_r[16]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[77]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.295																			
Hold: 'p0[altpl_0]sd1[p07]clk[0]'										34 -3.915 TopTopt0[AudDSP:dsp]addr_r[16]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[123]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.295																			
Hold: 'alt'era_reserved_tck'										35 -3.849 TopTopt0[AudRecorder:recorder]address_r[10]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[71]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.307																			
Hold: 'alt'era_reserved_tck'										36 -3.844 TopTopt0[AudRecorder:recorder]address_r[10]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[71]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.307																			
Hold: 'p0[altpl_0]sd1[p07]clk[1]'										37 -3.838 TopTopt0[AudDSP:dsp]addr_r[12]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[87]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.551																			
Hold: 'AUD_BCLK'										38 -3.838 TopTopt0[AudDSP:dsp]addr_r[6]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[86]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.551																			
Recovery: 'alt'era_reserved_tck'										39 -3.836 TopTopt0[AudDSP:dsp]addr_r[12]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[73]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.551																			
Removal: 'alt'era_reserved_tck'										40 -3.836 TopTopt0[AudDSP:dsp]addr_r[6]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[86]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.551																			
Minimum Pulse Width: 'CLOCK_50'										41 -3.808 TopTopt0[AudDSP:dsp]addr_r[0]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[70]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.522																			
Minimum Pulse Width: 'CLOCK2_50'										42 -3.808 TopTopt0[AudDSP:dsp]addr_r[0]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[70]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.522																			
Minimum Pulse Width: 'CLOCK3_50'										43 -3.807 TopTopt0[AudDSP:dsp]addr_r[5]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[85]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.525																			
Minimum Pulse Width: 'AUD_BCLK'										44 -3.806 TopTopt0[AudDSP:dsp]addr_r[5]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[85]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.525																			
Minimum Pulse Width: 'p0[altpl_0]sd1[p07]clk[1]'										45 -3.805 TopTopt0[AudRecorder:recorder]address_r[7]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[87]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.522																			
Minimum Pulse Width: 'p0[altpl_0]sd1[p07]clk[0]'										46 -3.806 TopTopt0[AudDSP:dsp]addr_r[7]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[87]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.522																			
Minimum Pulse Width: 'alt'era_reserved_tck'										47 -3.790 TopTopt0[AudDSP:dsp]addr_r[8]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[88]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.504																			
Minimum Pulse Width: 'p0[altpl_0]sd1[p07]clk[1]'										48 -3.788 TopTopt0[AudDSP:dsp]addr_r[8]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[88]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.504																			
Dataset Report										49 -3.785 TopTopt0[AudDSP:dsp]addr_r[14]										sid_signaltap:auto_signaltap_0laccq_trigger_in_reg[75]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.480																			
Metaheuristic Summary:										50 -3.785 TopTopt0[AudDSP:dsp]addr_r[14]										sid_signaltap:auto_signaltap_0laccq_data_in_reg[75]										AUD_BCLK										p0[altpl_0]sd1[p07]clk[0]										0.001										-2.480																			

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Table of Contents		Slow 1200mV OC Model Setup: 'AUD_BCLK'			
Flow Summary		Slack	From Node	To Node	Launch Clock
Flow Settings		-2.451	Debounce:deb2/neg_r	Top:top0 AudDSP:dsp data_next_r[5]	p0 atp0_0 sd1 p07 ck[0]
Flow Non-Default Global Settings		-2.398	Top:top0 dsp_start_r	Top:top0 AudDSP:dsp data_next_r[5]	AUD_BCLK
Flow Elapsed Time		-2.319	Debounce:deb2/neg_r	Top:top0 AudRecorder:recorder state_r_s_STOP	AUD_BCLK
Flow OS Summary		-2.317	Top:top0 dsp_start_r	Top:top0 AudDSP:dsp state_r_s_IDLE	AUD_BCLK
Flow Log		-2.290	Top:top0 rec_start_r	Top:top0 AudRecorder:recorder state_r_s_LEFT	AUD_BCLK
Analysis & Synthesis		-2.285	Top:top0 rec_start_r	Top:top0 AudRecorder:recorder state_r_s_IDLE	AUD_BCLK
Fitter		-2.242	Debounce:deb2/neg_r	Top:top0 AudDSP:dsp data_next_r[4]	AUD_BCLK
Flow Messages		-2.130	Top:top0 state_r_s_REC	Top:top0 AudDSP:dsp data_next_r[6]	AUD_BCLK
Flow Suppressed Messages		-2.106	Top:top0 state_r_s_REC	Top:top0 AudRecorder:recorder state_r_s_RIGHT	AUD_BCLK
Assembler		-2.099	Debounce:deb2/neg_r	Top:top0 AudDSP:dsp data_next_r[7]	AUD_BCLK
TimeQuest Timing Analyzer		-2.071	Top:top0 state_r_s_REC	Top:top0 AudDSP:dsp data_next_r[3]	AUD_BCLK
Summary		-2.038	Debounce:deb2/neg_r	Top:top0 AudDSP:dsp data_next_r[11]	AUD_BCLK
Parallel Completion		-2.032	Top:top0 state_r_s_REC	Top:top0 AudDSP:dsp data_next_r[12]	AUD_BCLK
SDC File List		-2.015	Top:top0 state_r_s_REC	Top:top0 AudDSP:dsp data_next_r[8]	AUD_BCLK
Clocks		-1.985	Top:top0 state_r_s_REC	Top:top0 AudDSP:dsp data_next_r[9]	AUD_BCLK
Slow 1200mV 85C Model		-1.971	Top:top0 state_r_s_REC	Top:top0 AudDSP:dsp data_next_r[1]	AUD_BCLK
Slow 1200mV OC Model		-1.966	Top:top0 state_r_s_REC	Top:top0 AudDSP:dsp data_next_r[10]	AUD_BCLK
Fmax Summary		-1.938	Top:top0 state_r_s_REC	Top:top0 AudDSP:dsp data_next_r[13]	AUD_BCLK
Setup Summary		-1.926	Top:top0 state_r_s_REC	Top:top0 AudDSP:dsp data_next_r[3]	AUD_BCLK
Hold Summary		-1.909	Top:top0 state_r_s_REC	Top:top0 AudDSP:dsp data_next_r[10]	AUD_BCLK
Recovery Summary		-1.879	Top:top0 state_r_s_REC	Top:top0 AudDSP:dsp data_next_r[13]	AUD_BCLK
Removal Summary		-1.825	Top:top0 state_r_s_REC	Top:top0 AudDSP:dsp data_next_r[12]	AUD_BCLK
Minimum Pulse Width Summary		-23.264	Top:top0 AudDSP:dsp data_next_r[0]	Top:top0 AudDSP:dsp data_next_r[14]	AUD_BCLK
Worst-Case Timing Paths		-23.236	Top:top0 AudDSP:dsp data_next_r[0]	Top:top0 AudDSP:dsp data_next_r[15]	AUD_BCLK
Setup: 'p0 atp0_0 sd1 p07 ck[0]'		-23.398	Top:top0 AudDSP:dsp data_next_r[0]	Top:top0 AudDSP:dsp data_next_r[12]	AUD_BCLK
Setup: 'p0 atp0_0 sd1 p07 ck[0]'		-23.412	Top:top0 AudDSP:dsp data_next_r[0]	Top:top0 AudDSP:dsp data_next_r[11]	AUD_BCLK
Setup: 'altera_reserved_tck'		-23.455	Top:top0 AudDSP:dsp data_next_r[2]	Top:top0 AudDSP:dsp data_next_r[14]	AUD_BCLK
Setup: 'p0 atp0_0 sd1 p07 ck[1]'		-23.477	Top:top0 AudDSP:dsp data_next_r[2]	Top:top0 AudDSP:dsp data_next_r[15]	AUD_BCLK
Setup: 'p0 atp0_0 sd1 p07 ck[1]'		-23.526	Top:top0 AudDSP:dsp data_next_r[0]	Top:top0 AudDSP:dsp data_next_r[13]	AUD_BCLK
Setup: 'AUD_BCLK'		-23.589	Top:top0 AudDSP:dsp data_next_r[2]	Top:top0 AudDSP:dsp data_next_r[12]	AUD_BCLK
Hold: 'p0 atp0_0 sd1 p07 ck[0]'		-23.591	Top:top0 AudDSP:dsp data_curr_r[1]	Top:top0 AudDSP:dsp data_next_r[14]	AUD_BCLK
Hold: 'altera_reserved_tck'		-23.603	Top:top0 AudDSP:dsp data_next_r[2]	Top:top0 AudDSP:dsp data_next_r[11]	AUD_BCLK
Hold: 'p0 atp0_0 sd1 p07 ck[1]'		-23.613	Top:top0 AudDSP:dsp data_curr_r[1]	Top:top0 AudDSP:dsp data_next_r[15]	AUD_BCLK
Hold: 'AUD_BCLK'		-23.622	Top:top0 AudDSP:dsp data_next_r[0]	Top:top0 AudDSP:dsp data_next_r[13]	AUD_BCLK
Recovery: 'altera_reserved_tck'		-23.637	Top:top0 AudDSP:dsp data_next_r[0]	Top:top0 AudDSP:dsp data_next_r[10]	AUD_BCLK
Removal: 'altera_reserved_tck'		-23.652	Top:top0 AudDSP:dsp data_curr_r[0]	Top:top0 AudDSP:dsp data_next_r[14]	AUD_BCLK
Minimum Pulse Width: 'CLOCK_50'		-23.655	Top:top0 AudDSP:dsp data_next_r[0]	Top:top0 AudDSP:dsp data_next_r[8]	AUD_BCLK
Minimum Pulse Width: 'CLOCK_50'		-23.674	Top:top0 AudDSP:dsp data_curr_r[0]	Top:top0 AudDSP:dsp data_next_r[15]	AUD_BCLK
Minimum Pulse Width: 'CLOCK_50'		-23.709	Top:top0 AudDSP:dsp data_curr_r[2]	Top:top0 AudDSP:dsp data_next_r[14]	AUD_BCLK
Minimum Pulse Width: 'AUD_BCLK'		-23.717	Top:top0 AudDSP:dsp data_next_r[2]	Top:top0 AudDSP:dsp data_next_r[13]	AUD_BCLK
Minimum Pulse Width: 'p0 atp0_0 sd1 p07 ck[0]'		-23.725	Top:top0 AudDSP:dsp data_curr_r[1]	Top:top0 AudDSP:dsp data_next_r[12]	AUD_BCLK
Minimum Pulse Width: 'altera_reserved_tck'		-23.731	Top:top0 AudDSP:dsp data_curr_r[2]	Top:top0 AudDSP:dsp data_next_r[15]	AUD_BCLK
Minimum Pulse Width: 'p0 atp0_0 sd1 p07 ck[1]'		-23.739	Top:top0 AudDSP:dsp data_next_r[1]	Top:top0 AudDSP:dsp data_next_r[11]	AUD_BCLK
Minimum Pulse Width: 'p0 atp0_0 sd1 p07 ck[1]'		-23.745	Top:top0 AudDSP:dsp data_next_r[0]	Top:top0 AudDSP:dsp data_next_r[8]	AUD_BCLK
Minimum Pulse Width: 'p0 atp0_0 sd1 p07 ck[1]'		-23.752	Top:top0 AudDSP:dsp data_next_r[0]	Top:top0 AudDSP:dsp data_next_r[14]	AUD_BCLK
Minimum Pulse Width: 'p0 atp0_0 sd1 p07 ck[1]'		-23.760	Top:top0 AudDSP:dsp data_next_r[0]	Top:top0 AudDSP:dsp data_next_r[9]	AUD_BCLK
Minimum Pulse Width: 'p0 atp0_0 sd1 p07 ck[1]'		-23.774	Top:top0 AudDSP:dsp data_curr_r[3]	Top:top0 AudDSP:dsp data_next_r[15]	AUD_BCLK

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Table of Contents		Slow 1200mV OC Model Setup: 'AUD_BCLK'			
Flow Summary		Slack	From Node	To Node	Launch Clock
Flow Settings		51 23.786	Top:top0 AudDSP:dsp data_curr_r[0]	Top:top0 AudDSP:dsp data_next_r[12]	AUD_BCLK
Flow Non-Default Global Settings		52 23.800	Top:top0 AudDSP:dsp data_curr_r[0]	Top:top0 AudDSP:dsp data_next_r[11]	AUD_BCLK
Flow Elapsed Time		53 23.812	Top:top0 AudDSP:dsp data_next_r[0]	Top:top0 AudDSP:dsp data_next_r[15]	AUD_BCLK
Flow OS Summary		54 23.814	Top:top0 AudDSP:dsp data_next_r[2]	Top:top0 AudDSP:dsp data_next_r[14]	AUD_BCLK
Flow Log		55 23.814	Top:top0 AudDSP:dsp data_curr_r[4]	Top:top0 AudDSP:dsp data_next_r[14]	AUD_BCLK
Analysis & Synthesis		56 23.828	Top:top0 AudDSP:dsp data_next_r[2]	Top:top0 AudDSP:dsp data_next_r[10]	AUD_BCLK
Fitter		57 23.836	Top:top0 AudDSP:dsp data_curr_r[4]	Top:top0 AudDSP:dsp data_next_r[15]	AUD_BCLK
Flow Messages		58 23.843	Top:top0 AudDSP:dsp data_curr_r[2]	Top:top0 AudDSP:dsp data_next_r[12]	AUD_BCLK
Flow Suppressed Messages		59 23.846	Top:top0 AudDSP:dsp data_next_r[2]	Top:top0 AudDSP:dsp data_next_r[8]	AUD_BCLK
Assembler		60 23.853	Top:top0 AudDSP:dsp data_curr_r[1]	Top:top0 AudDSP:dsp data_next_r[13]	AUD_BCLK
TimeQuest Timing Analyzer		61 23.857	Top:top0 AudDSP:dsp data_curr_r[2]	Top:top0 AudDSP:dsp data_next_r[11]	AUD_BCLK
Summary		62 23.879	Top:top0 AudDSP:dsp data_curr_r[6]	Top:top0 AudDSP:dsp data_next_r[14]	AUD_BCLK
Parallel Completion		63 23.886	Top:top0 AudDSP:dsp data_curr_r[3]	Top:top0 AudDSP:dsp data_next_r[12]	AUD_BCLK
SDC File List		64 23.892	Top:top0 AudDSP:dsp data_next_r[0]	Top:top0 AudDSP:dsp data_next_r[4]	AUD_BCLK
Clocks		65 23.900	Top:top0 AudDSP:dsp data_curr_r[3]	Top:top0 AudDSP:dsp data_next_r[11]	AUD_BCLK
Slow 1200mV 85C Model		66 23.901	Top:top0 AudDSP:dsp data_curr_r[6]	Top:top0 AudDSP:dsp data_next_r[15]	AUD_BCLK
Slow 1200mV OC Model		67 23.909	Top:top0 AudDSP:dsp data_curr_r[5]	Top:top0 AudDSP:dsp data_next_r[14]	AUD_BCLK
Fmax Summary		68 23.914	Top:top0 AudDSP:dsp data_curr_r[0]	Top:top0 AudDSP:dsp data_next_r[13]	AUD_BCLK
Setup Summary		69 23.931	Top:top0 AudDSP:dsp data_curr_r[5]	Top:top0 AudDSP:dsp data_next_r[15]	AUD_BCLK
Hold Summary		70 23.936	Top:top0 AudDSP:dsp data_next_r[2]	Top:top0 AudDSP:dsp data_next_r[6]	AUD_BCLK
Recovery Summary		71 23.948	Top:top0 AudDSP:dsp data_curr_r[4]	Top:top0 AudDSP:dsp data_next_r[12]	AUD_BCLK
Removal Summary		72 23.950	Top:top0 AudDSP:dsp data_curr_r[1]	Top:top0 AudDSP:dsp data_next_r[7]	AUD_BCLK
Minimum Pulse Width Summary		73 23.951	Top:top0 AudDSP:dsp data_next_r[2]	Top:top0 AudDSP:dsp data_next_r[9]	AUD_BCLK
Worst-Case Timing Paths		74 23.962	Top:top0 AudDSP:dsp data_curr_r[4]	Top:top0 AudDSP:dsp data_next_r[11]	AUD_BCLK
Setup: 'p0 atp0_0 sd1 p07 ck[0]'		75 23.964	Top:top0 AudDSP:dsp data_curr_r[1]	Top:top0 AudDSP:dsp data_next_r[10]	AUD_BCLK
Setup: 'altera_reserved_tck'		76 23.971	Top:top0 AudDSP:dsp data_curr_r[7]	Top:top0 AudDSP:dsp data_next_r[14]	AUD_BCLK
Setup: 'p0 atp0_0 sd1 p07 ck[1]'		77 23.971	Top:top0 AudDSP:dsp data_curr_r[2]	Top:top0 AudDSP:dsp data_next_r[13]	AUD_BCLK
Setup: 'p0 atp0_0 sd1 p07 ck[0]'		78 23.981	Top:top0 AudDSP:dsp data_next_r[0]	Top:top0 AudDSP:dsp data_next_r[2]	AUD_BCLK
Setup: 'altera_reserved_tck'		79 23.982	Top:top0 AudDSP:dsp data_curr_r[1]	Top:top0 AudDSP:dsp data_next_r[8]	AUD_BCLK
Setup: 'p0 atp0_0 sd1 p07 ck[1]'		80 23.993	Top:top0 AudDSP:dsp data_curr_r[7]	Top:top0 AudDSP:dsp data_next_r[15]	AUD_BCLK
Setup: 'AUD_BCLK'		81 24.004	Top:top0 AudDSP:dsp data_next_r[2]	Top:top0 AudDSP:dsp data_next_r[15]	AUD_BCLK
Hold: 'altera_reserved_tck'		82 24.011	Top:top0 AudDSP:dsp data_curr_r[0]	Top:top0 AudDSP:dsp data_next_r[7]	AUD_BCLK
Hold: 'p0 atp0_0 sd1 p07 ck[1]'		83 24.013	Top:top0 AudDSP:dsp data_curr_r[6]	Top:top0 AudDSP:dsp data_next_r[12]	AUD_BCLK
Hold: 'altera_reserved_tck'		84 24.014	Top:top0 AudDSP:dsp data_curr_r[3]	Top:top0 AudDSP:dsp data_next_r[10]	AUD_BCLK
Hold: 'p0 atp0_0 sd1 p07 ck[1]'		85 24.025	Top:top0 AudDSP:dsp data_curr_r[6]	Top:top0 AudDSP:dsp data_next_r[11]	AUD_BCLK
Hold: 'AUD_BCLK'		86 24.027	Top:top0 AudDSP:dsp data_curr_r[0]	Top:top0 AudDSP:dsp data_next_r[8]	AUD_BCLK
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Minimum Pulse Width: 'CLOCK_50'		90 24.072	Top:top0 AudDSP:dsp data_next_r[1]	Top:top0 AudDSP:dsp data_next_r[6]	AUD_BCLK
Minimum Pulse Width: 'AUD_BCLK'		91 24.076	Top:top0 AudDSP:dsp data_curr_r[4]	Top:top0 AudDSP:dsp data_next_r[13]	AUD_BCLK
Minimum Pulse Width: 'p0 atp0_0 sd1 p07 ck[0]'		92 24.079	Top:top0 AudDSP:dsp data_next_r[13]	Top:top0 AudDSP:dsp data_next_r[14]	AUD_BCLK
Minimum Pulse Width: 'altera_reserved_tck'		93 24.082	Top:top0 AudDSP:dsp data_curr_r[2]	Top:top0 AudDSP:dsp data_next_r[10]	AUD_BCLK
Minimum Pulse Width: 'p0 atp0_0 sd1 p07 ck[1]'		94 24.083	Top:top0 AudDSP:dsp data_next_r[2]	Top:top0 AudDSP:dsp data_next_r[9]	AUD_BCLK
Minimum Pulse Width: 'altera_reserved_tck'		95 24.087	Top:top0 AudDSP:dsp data_curr_r[1]	Top:top0 AudDSP:dsp data_next_r[14]	AUD_BCLK
Minimum Pulse Width: 'p0 atp0_0 sd1 p07 ck[1]'		96 24.093	Top:top0 AudDSP:dsp data_next_r[8]	Top:top0 AudDSP:dsp data_next_r[9]	AUD_BCLK
Minimum Pulse Width: 'p0 atp0_0 sd1 p07 ck[1]'		97 24.100	Top:top0 AudDSP:dsp data_curr_r[2]	Top:top0 AudDSP:dsp data_next_r[8]	AUD_BCLK
Minimum Pulse Width: 'p0 atp0_0 sd1 p07 ck[1]'		98 24.101	Top:top0 AudDSP:dsp data_next_r[13]	Top:top0 AudDSP:dsp data_next_r[15]	AUD_BCLK
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51 0.355

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52 0.355

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53 0.355

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54 0.355

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: ager:sd1_buffer_manager_inst/buffer_manager_done

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: ager:sd1_buffer_manager_inst/buffer_manager

55 0.355

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56 0.355

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: r_manager_inst/buffer_manager_isbuffer_wrapped

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57 0.355

DebounceDeb1|counter_r[12]

DebounceDeb1|counter_r[12]

58 0.355

DebounceDeb1|io_debounced_r

DebounceDeb1|io_debounced_r

59 0.355

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DebounceDeb1|counter_r[1]

60 0.355

Top:top0|rc_done_r

Top:top0|rc_done_r

61 0.355

Top:top0|state_r,S_PLAY

Top:top0|state_r,S_PLAY

62 0.356

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63 0.358

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64 0.360

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65 0.360

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66 0.361

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67 0.362

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68 0.363

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69 0.365

DebounceDeb2|counter_r[0]

DebounceDeb2|counter_r[0]

70 0.366

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sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: i:auo_generated/ram_block1a18-por

71 0.366

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sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm1:74

72 0.366

DebounceDeb2|counter_r[0]

DebounceDeb2|counter_r[0]

73 0.367

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: pb:sd1_signalnat_body/lacq_data_in_pipe_reg[3][57]

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: i:auo_generated/ram_block1a54-por

74 0.367

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: pb:sd1_signalnat_body/lacq_data_in_pipe_reg[3][39]

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: i:auo_generated/ram_block1a36-por

75 0.368

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: pb:sd1_signalnat_body/lacq_data_in_pipe_reg[3][47]

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: i:auo_generated/ram_block1a36-por

76 0.371

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: pb:sd1_signalnat_body/lacq_data_in_pipe_reg[3][45]

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77 0.373

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78 0.373

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79 0.374

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80 0.376

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: pb:sd1_signalnat_body/lacq_data_in_pipe_reg[3][63]

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81 0.377

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82 0.379

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83 0.379

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm2:52sm1|hoffff

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm2:52

84 0.380

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sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: i:auo_generated/ram_block1a18-por

85 0.380

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm3:49sm1|hoffff

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm3:49

86 0.380

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm3:61sm1|hoffff

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm3:61

87 0.380

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm3:135sm1|hoffff

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm3:135

88 0.380

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm3:77sm1|hoffff

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm3:77

89 0.381

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm4:46sm1|hoffff

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm4:46

90 0.381

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm4:7sm1|hoffff

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm4:7

91 0.381

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm4:4sm1|hoffff

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm4:4

92 0.381

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sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: pb:sd1_signalnat_body/lacq_data_in_p

93 0.381

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm2:28sm1|hoffff

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm2:28

94 0.381

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm2:26sm1|hoffff

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95 0.381

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm2:7sm1|hoffff

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm2:7

96 0.381

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm2:70sm1|hoffff

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm2:70

97 0.381

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm2:69sm1|hoffff

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm2:69

98 0.381

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm2:121sm1|hoffff

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm2:121

99 0.382

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm2:11sm1|hoffff

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: sbmpg_pipeline_less_than_two:sm2:11

99 0.382

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: pb:sd1_signalnat_body/lacq_data_in_p

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: pb:sd1_signalnat_body/lacq_data_in_p

100 0.382

sd1_signalnat:auto_signalnat_0lsd_signalnat_mpl: n_sbmop_pipeline_less_than_two:sm2:20sm1|hoffff

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1	p10 altpll_0 sd1 pll7 clk[0]	-2.861	-127.119
2	AUD_BCLK	-1.562	-29.044
3	altera_reserved_tck	46.792	0.000
4	p10 altpll_0 sd1 pll7 clk[1]	81.005	0.000

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	Clock	Slack	End Point TNS
1	p10 altpll_0 sd1 pll7 clk[0]	-0.175	-0.346
2	altera_reserved_tck	0.181	0.000
3	p10 altpll_0 sd1 pll7 clk[1]	0.184	0.000
4	AUD_BCLK	0.201	0.000

Multicorner Timing Analysis Summary						
	Clock	Setup	Hold	Recovery	Removal	Minimum Pulse Width
1	Worst-case Slack	-5.432	-0.371	48.398	0.489	9.400
1	AUD_BCLK	-2.835	0.201	N/A	N/A	40.561
2	CLOCK2_50	N/A	N/A	N/A	N/A	16.000
3	CLOCK3_50	N/A	N/A	N/A	N/A	16.000
4	CLOCK_50	N/A	N/A	N/A	N/A	9.400
5	altera_reserved_tck	43.021	0.181	48.398	0.489	49.283
6	p10[altpll_0]sd1[pll7]clk[0]	-5.432	-0.371	N/A	N/A	41.354
7	p10[altpll_0]sd1[pll7]clk[1]	78.961	0.184	N/A	N/A	4999.708
2	Design-wide THS	-299.503	-0.738	0.0	0.0	0.0
1	AUD_BCLK	-53.456	0.000	N/A	N/A	0.000
2	CLOCK2_50	N/A	N/A	N/A	N/A	0.000
3	CLOCK3_50	N/A	N/A	N/A	N/A	0.000
4	CLOCK_50	N/A	N/A	N/A	N/A	0.000
5	altera_reserved_tck	0.000	0.000	0.000	0.000	0.000
6	p10[altpll_0]sd1[pll7]clk[0]	-246.047	-0.738	N/A	N/A	0.000
7	p10[altpll_0]sd1[pll7]clk[1]	0.000	0.000	N/A	N/A	0.000

6. 心得

這次的lab比起前兩次的複雜非常多，因此在這次的debug過程非常的漫長，如果把整個程式都統合起來之後才一起進行測試很難知道到底是哪一部分出問題，因此我們有用了LED(綠)來顯示目前狀態來確保到底有沒有進到下一個state，此外我們還學習利用quartus中的SignalTap II Logic Analyzer File 來去看各個register與input等等在操作過程中的值是多少，來幫助debug